

Right Anodal tDCS Modulates Alerting in Chronic Post-Stroke Aphasia

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Background

Attention deficits are common in post-stroke aphasia and may limit language recovery¹. Attention comprises three partially independent networks (Alerting, Orienting, and Executive Control)² that can be differentially affected in aphasia³. Because the right hemisphere is critical for alerting and sustained attention⁴, it may represent a promising neuromodulation target. However, most tDCS studies in aphasia have focused on left-hemisphere language regions, leaving the effects of right-anodal stimulation on attentional networks and language outcomes largely unexplored^{5,6}.

Objectives

- To investigate whether right-hemisphere anodal tDCS, combined with donepezil and intensive Language Action Therapy (ILAT), modulates alerting, orienting, and executive control networks.
- To determine whether changes in attentional networks are associated with language and communication outcomes in chronic post-stroke aphasia.

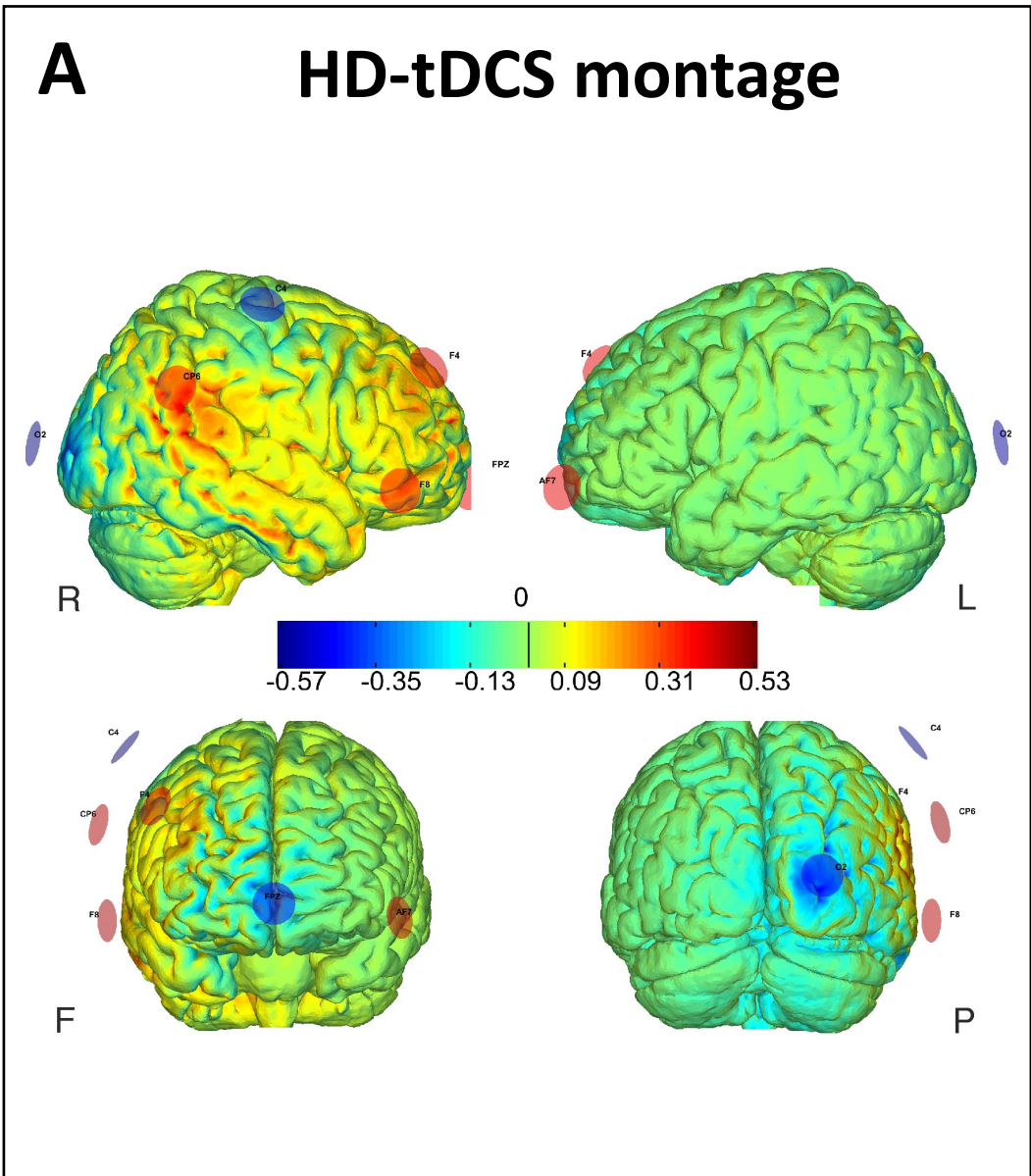
Methods

Sample and design. Twenty adults with chronic post-stroke aphasia, all receiving donepezil 10 mg/day were randomized to anodal tDCS + ILAT (n=10) or sham tDCS + ILAT (n=10).

Stimulation. HD-tDCS (7-electrode montage; principal anode over right perisylvian sites F4/F8/CP6; 4.0 mA total current; **A**) was delivered in ten 25-min sessions over two weeks (10 days), concurrent with ILAT; sham used an identical montage with brief ramp-up/down only.

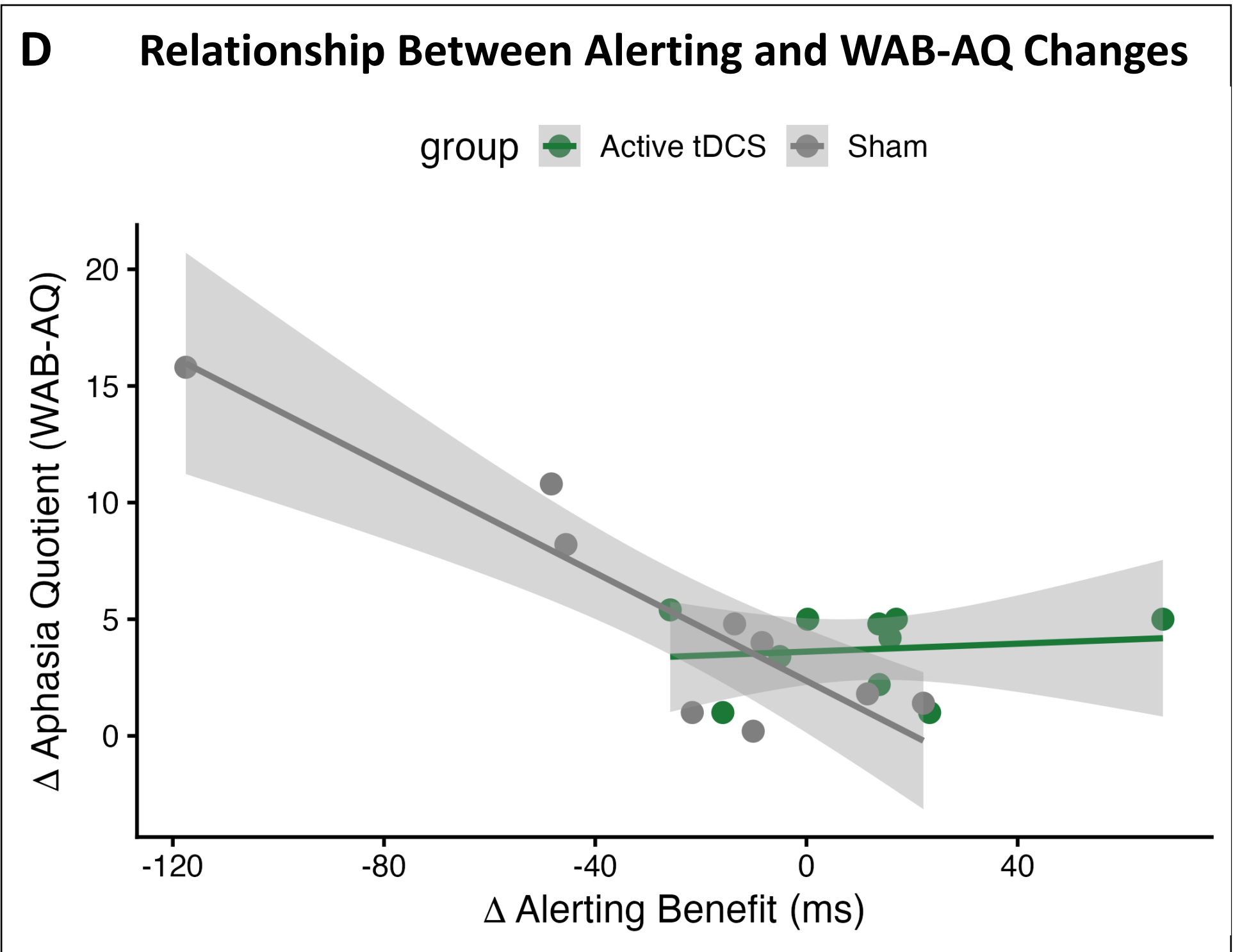
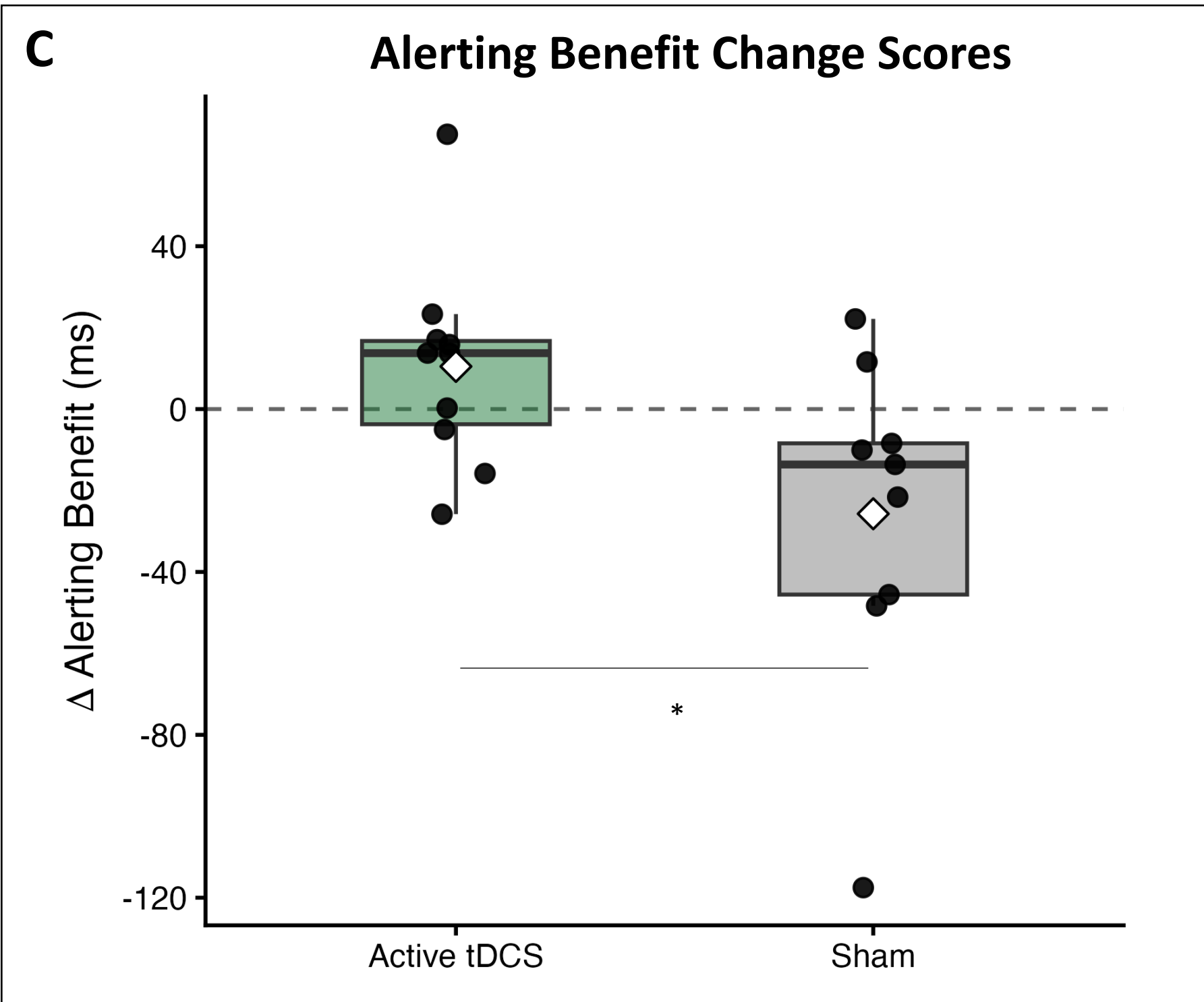
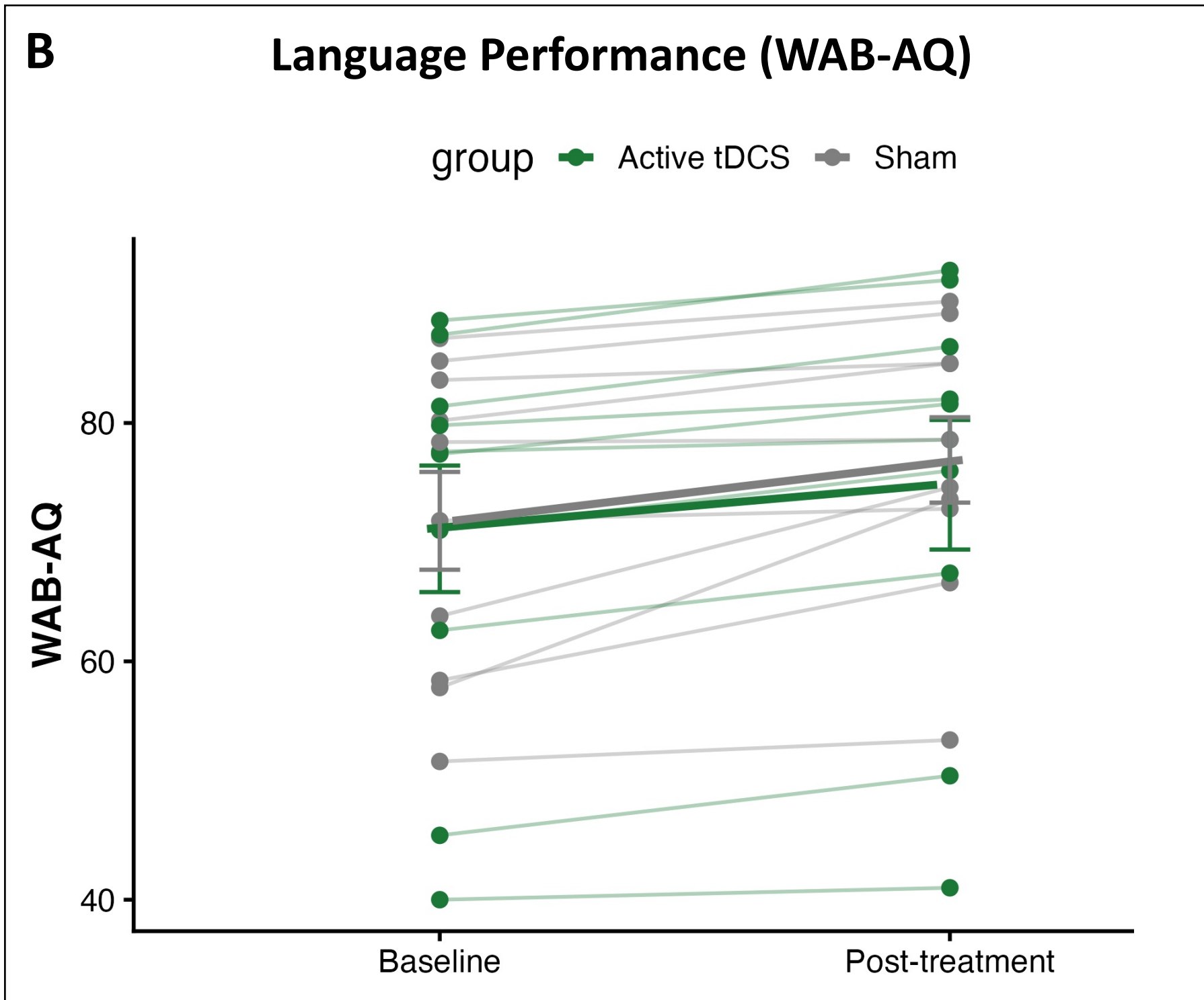
Assessment. Attention: Attention Network Test for Interactions (ANTI), language (WAB-AQ), functional communication (CAL), quality of life (SAQOL-39) were assessed pre- and post-intervention.

Analysis. Pre-post group differences via t-tests / Mann-Whitney U and Linear Mixed Models (adjusted for age, lesion volume, time post-stroke); moderation analysis testing whether tDCS group altered the relationship between attentional-network change and WAB-AQ gains.



Results

- Language, communication and quality of life.** Both groups improved significantly from baseline to post-treatment in language (WAB-AQ, $p=.004$; **B**), communication (CAL, $p=.020$), and quality of life (SAQOL-39, $p=.002$), with no significant between-group differences (all $p>.14$).
- Attention.** The only group difference emerged in the Alerting network: alerting benefit increased after anodal tDCS ($\Delta=+10.5$ ms) but declined with sham tDCS ($\Delta=-25.7$ ms; $d=1.07$, $p=.041$; **C**).
- Alerting-Language relationship.** Moderation analysis showed that changes in Alerting were differently associated with language gains across groups ($p=.001$, $R^2=.78$; **D**). In the sham group, greater loss of alerting benefit was associated with larger WAB-AQ gains ($\rho=-0.68$, $p=.042$). This relationship was absent after anodal tDCS ($\rho=-0.06$, $p=.866$). Concurrent gains in attention and language were more frequent with anodal tDCS (7/10 vs. 2/9 in sham).



Conclusions

- Right-hemisphere anodal tDCS did not confer additional language or communication benefits beyond those of donepezil and ILAT.
- Its effect may instead relate to the allocation of attentional resources during recovery. In the sham group, greater language gains were associated with larger declines in auditory alerting, suggesting a potential language-attention trade-off during intensive therapy. This association was absent after anodal tDCS. This pattern is consistent with the idea that ILAT may partially rely on domain-general attentional resources and that right-hemisphere stimulation may support alerting capacity without competing with language gains.
- These preliminary findings provide a theoretically grounded hypothesis for future adequately powered trials investigating the role of right-hemisphere stimulation in chronic aphasia rehabilitation.

References

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